Renal

"But I know all about love already. I know precious little still about kidneys."

—Aldous Huxley, *Antic Hay*

"This too shall pass. Just like a kidney stone."

—Hunter Madsen

"I drink too much. The last time I gave a urine sample it had an olive in it."

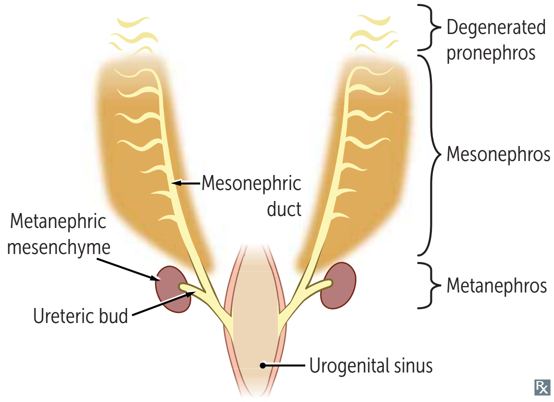
—Rodney Dangerfield

▶ RENAL—EMBRYOLOGY

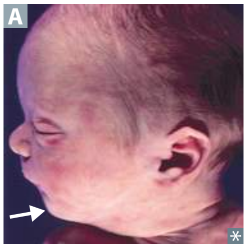
Kidney embryology

Pronephros—week 4; then degenerates.
Mesonephros—functions as interim kidney for 1st trimester; later contributes to male genital system.
Metanephros—permanent; first appears in 5th week of gestation; nephrogenesis continues through weeks 32–36 of gestation.
- Ureteric bud—derived from caudal end of mesonephric duct; gives rise to ureter, pelvises, calyces, collecting ducts; fully canalized by 10th week
- Metanephric mesenchyme (ie, metanephric blastema)—ureteric bud interacts with this tissue; interaction induces differentiation and formation of glomerulus through to distal convoluted tubule (DCT)
- Aberrant interaction between these 2 tissues may result in several congenital malformations of the kidney

Ureteropelvic junction—last to canalize → most common site of obstruction (can be detected on prenatal ultrasound as hydronephrosis).



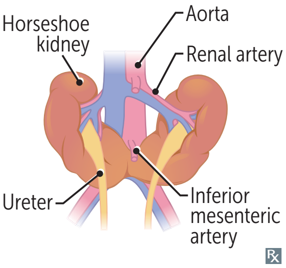
Potter sequence (syndrome)



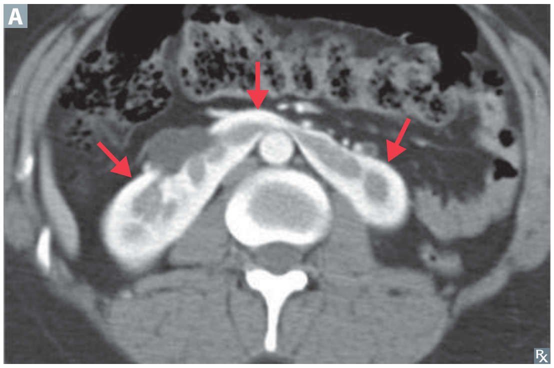
Oligohydramnios → compression of developing fetus → limb deformities, facial anomalies (eg, low-set ears and retrognathia A, flattened nose), compression of chest and lack of amniotic fluid aspiration into fetal lungs → pulmonary hypoplasia (cause of death).
Causes include ARPKD, obstructive uropathy (eg, posterior urethral valves), bilateral renal agenesis, chronic placental insufficiency.

Babies who can't "Pee" in utero develop Potter sequence.
POTTER sequence associated with:
Pulmonary hypoplasia
Oligohydramnios (trigger)
Twisted face
Twisted skin
Extremity defects
Renal failure (in utero)

Horseshoe kidney



Inferior poles of both kidneys fuse abnormally A. As they ascend from pelvis during fetal development, horseshoe kidneys get trapped under inferior mesenteric artery and remain low in the abdomen. Kidneys function normally. Associated with hydronephrosis (eg, ureteropelvic junction obstruction), renal stones, infection, chromosomal aneuploidy syndromes (eg, Turner syndrome; trisomies 13, 18, 21), and rarely renal cancer.



Congenital solitary functioning kidney

Condition of being born with only one functioning kidney. Majority asymptomatic with compensatory hypertrophy of contralateral kidney, but anomalies in contralateral kidney are common. Often diagnosed prenatally via ultrasound.

Unilateral renal agenesis

Ureteric bud fails to develop and induce differentiation of metanephric mesenchyme → complete absence of kidney and ureter.

Multicystic dysplastic kidney

Ureteric bud fails to induce differentiation of metanephric mesenchyme → nonfunctional kidney consisting of cysts and connective tissue. Predominantly nonhereditary and usually unilateral; bilateral leads to Potter sequence.

Duplex collecting system

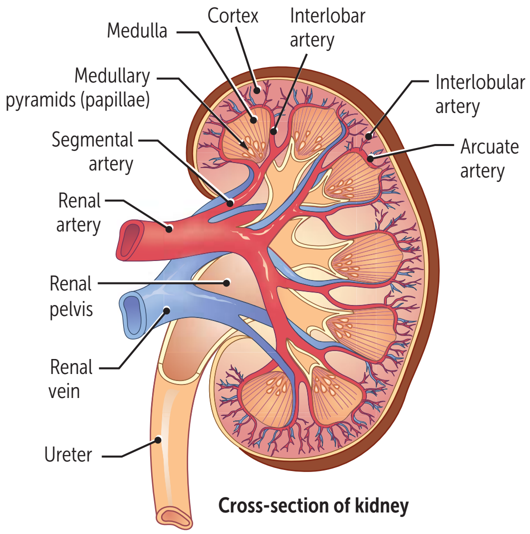
Bifurcation of ureteric bud before it enters the metanephric blastema creates a Y-shaped bifid ureter. Duplex collecting system can alternatively occur through two ureteric buds reaching and interacting with metanephric blastema. Strongly associated with vesicoureteral reflux and/or ureteral obstruction, ↑ risk for UTIs.

Posterior urethral valves

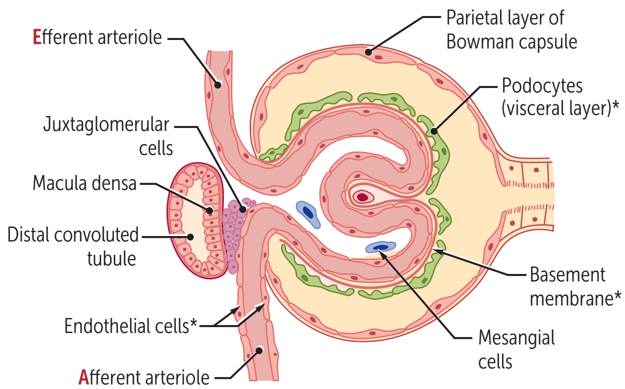
Membrane remnant in the posterior urethra in males; its persistence can lead to urethral obstruction. Can be diagnosed prenatally by hydronephrosis and dilated or thick-walled bladder on ultrasound. Most common cause of bladder outlet obstruction in male infants.

▸ RENAL—ANATOMY

Kidney anatomy and glomerular structure



Cross-section of kidney



*Components of glomerular filtration barrier.

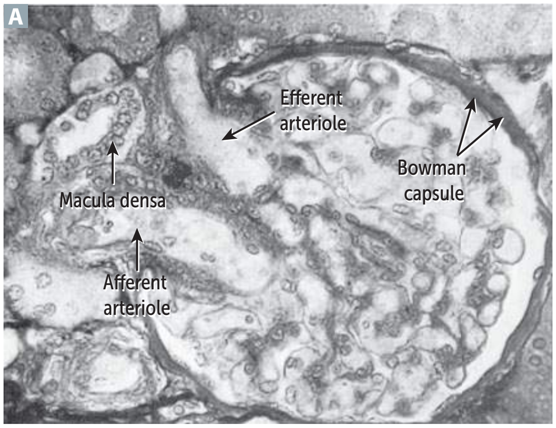
Cross-section of glomerulus A

Left kidney is taken during donor transplantation because it has a longer renal vein.

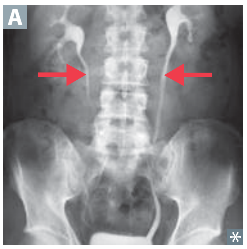
Afferent = **A**rriving.

Efferent = **E**xiting.

Renal blood flow: renal artery → segmental artery → interlobar artery → arcuate artery → interlobular artery → afferent arteriole → glomerulus → efferent arteriole → vasa recta/peritubular capillaries → venous outflow.



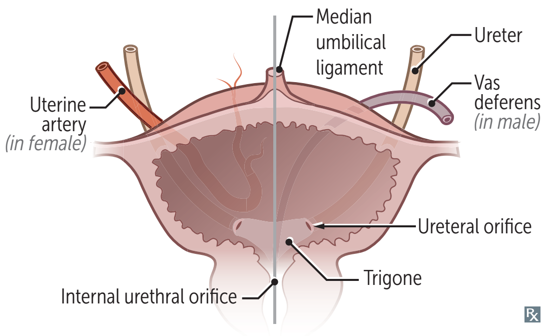
Course of ureters



Ureters A pass **under** uterine artery or **under** vas deferens (retroperitoneal).

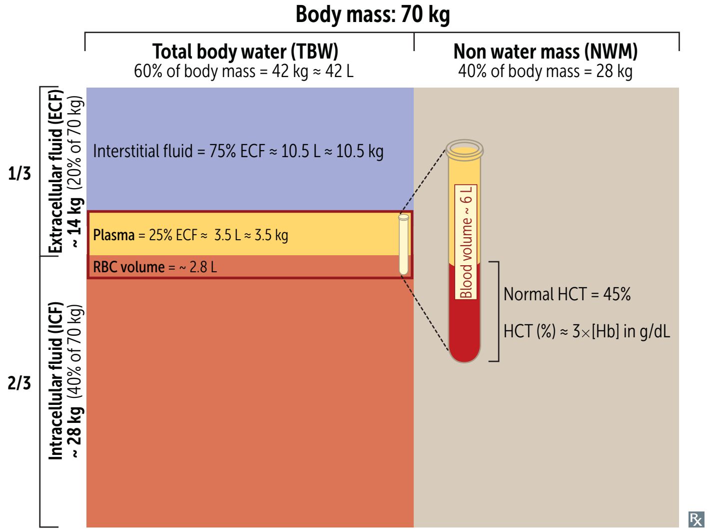
Gynecologic procedures (eg, ligation of uterine or ovarian vessels) may damage ureter → ureteral obstruction or leak.

"Water (ureters) **under** the bridge (uterine artery or vas deferens)."



▶ RENAL—PHYSIOLOGY

Fluid compartments



HIKIN': **HI**gh **K⁺ IN**tracellularly.

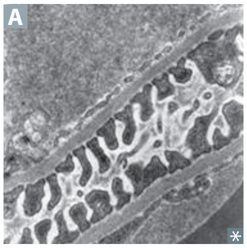
60–40–20 rule (% of body weight for average person):
- 60% total body water
- 40% ICF
- 20% ECF

Plasma volume can be measured by radiolabeling albumin.

Extracellular volume can be measured by inulin or mannitol.

Osmolality = 285–295 mOsm/kg H_2O.

Glomerular filtration barrier



Responsible for filtration of plasma according to size and charge selectivity.

Composed of:
- Fenestrated capillary endothelium
- Basement membrane with type IV collagen chains and heparan sulfate
- Epithelial layer consisting of podocyte foot processes A

Charge barrier—all 3 layers contain ⊖ charged glycoproteins preventing ⊕ charged molecule entry (eg, albumin).

Size barrier—fenestrated capillary epithelium (prevent entry of > 100 nm molecules/blood cells); podocyte foot processes interpose with basement membrane; slit diaphragm (prevent entry of molecules > 50–60 nm).

Renal clearance

$C_x = U_xV/P_x$ = volume of plasma from which the substance is completely cleared per unit time.
If $C_x <$ GFR: net tubular reabsorption of X.
If $C_x >$ GFR: net tubular secretion of X.
If $C_x =$ GFR: no net secretion or reabsorption.

C_x = clearance of X (mL/min).
U_x = urine concentration of X (eg, mg/mL).
P_x = plasma concentration of X (eg, mg/mL).
V = urine flow rate (mL/min).

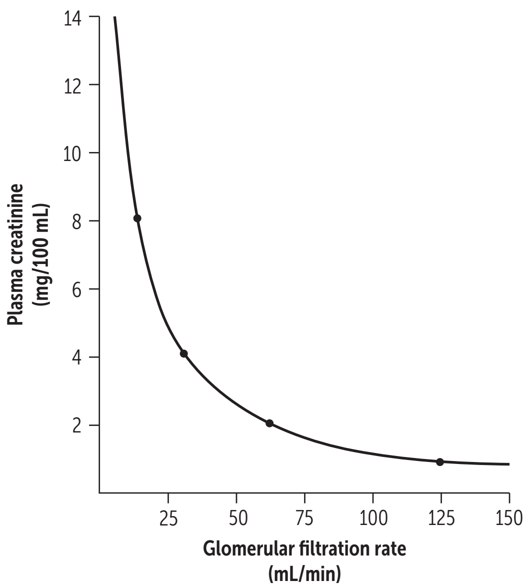
Glomerular filtration rate

Inulin clearance can be used to calculate GFR because it is freely filtered and is neither reabsorbed nor secreted.

$$\text{GFR} = U_{\text{inulin}} \times V/P_{\text{inulin}} = C_{\text{inulin}}$$
$$= K_f\,[(P_{GC} - P_{BS}) - (\pi_{GC} - \pi_{BS})]$$

(GC = glomerular capillary; BS = Bowman space.) π_{BS} normally equals zero; K_f = filtration constant.

Normal GFR ≈ 100 mL/min.
Creatinine clearance is an approximate measure of GFR. Slightly overestimates GFR because creatinine is moderately secreted by renal tubules.
Incremental reductions in GFR define the stages of chronic kidney disease.



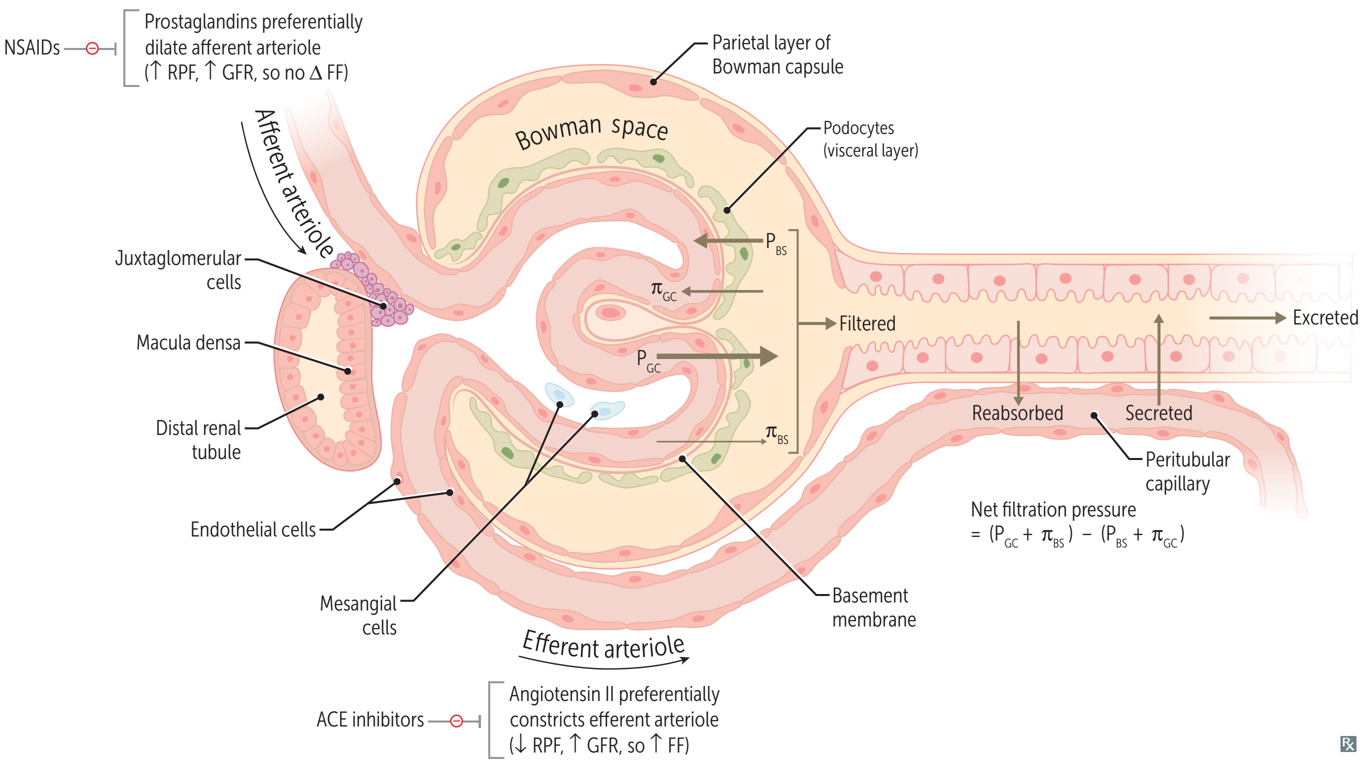
Effective renal plasma flow

Effective renal plasma flow (eRPF) can be estimated using *para*-aminohippuric acid (PAH) clearance. Between filtration and secretion, there is nearly 100% excretion of all PAH that enters the kidney.
$\text{eRPF} = U_{PAH} \times V/P_{PAH} = C_{PAH}$.
Renal blood flow (RBF) = RPF/(1 – Hct).
Plasma = 1 – hematocrit.
eRPF underestimates true renal plasma flow (RPF) slightly.

Filtration

Filtration fraction (FF) = GFR/RPF.
Normal FF = 20%.
Filtered load (mg/min) = GFR (mL/min) × plasma concentration (mg/mL).

GFR can be estimated with creatinine clearance.
RPF is best estimated with PAH clearance.
Prostaglandins **D**ilate **A**fferent arteriole (**PDA**)
ACE inhibitors **C**onstrict **E**fferent arteriole (**ACE**)



Changes in glomerular dynamics

Effect	GFR	RPF	FF (GFR/RPF)
Afferent arteriole constriction	↓	↓	—
Efferent arteriole constriction	↑	↓	↑
↑ plasma protein concentration	↓	—	↓
↓ plasma protein concentration	↑	—	↑
Constriction of ureter	↓	—	↓
Dehydration	↓	↓↓	↑

Calculation of reabsorption and secretion rate

Filtered load = GFR × P_x.
Excretion rate = V × U_x.
Reabsorption rate = filtered – excreted.
Secretion rate = excreted – filtered.

$$Fe_{Na} = \frac{Na^+ \text{ excreted}}{Na^+ \text{ filtered}} = \frac{V \times U_{Na}}{GFR\left(U_{Cr} \times \frac{V}{P_{Cr}}\right) \times P_{Na}} = \frac{P_{Cr} \times U_{Na}}{U_{Cr} \times P_{Na}}$$

Glucose clearance

Glucose at a normal plasma level (range 60–120 mg/dL) is completely reabsorbed in proximal convoluted tubule (PCT) by Na^+/glucose cotransport.

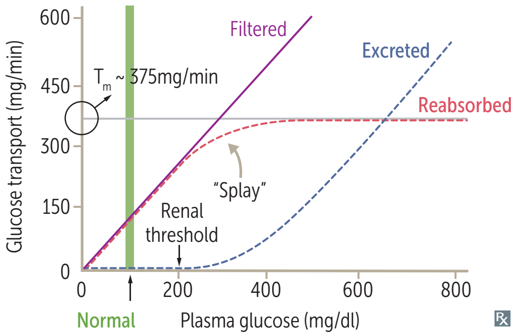
In adults, at plasma glucose of ~ 200 mg/dL, glucosuria begins (threshold). At rate of ~ 375 mg/min, all transporters are fully saturated (T_m).

Normal pregnancy may decrease ability of PCT to reabsorb glucose and amino acids → glucosuria and aminoaciduria.

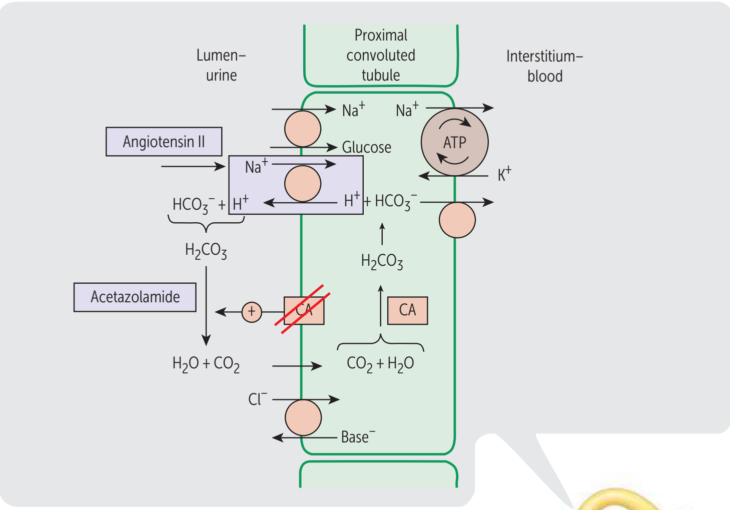
Sodium-glucose cotransporter 2 (SGLT2) inhibitors (eg, -flozin drugs) permit glucosuria at plasma concentrations < 200 mg/dL.

Glucosuria is an important clinical clue to diabetes mellitus.

Splay is the region of substance clearance between threshold and T_m; due to the heterogeneity of nephrons.



Nephron physiology



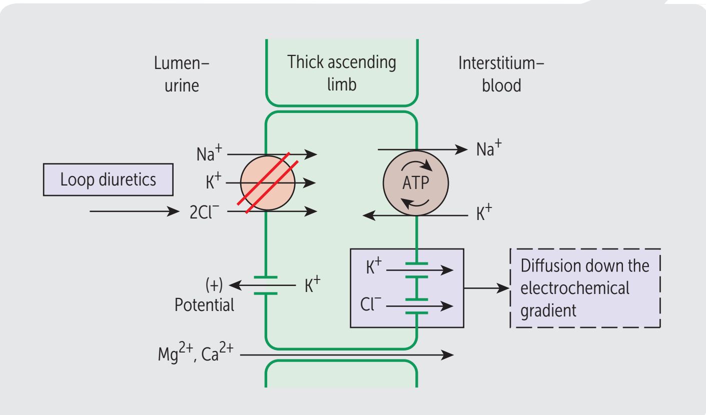
Early PCT—contains brush border. Reabsorbs all glucose and amino acids and most HCO_3^-, Na^+, Cl^-, PO_4^{3-}, K^+, H_2O, and uric acid. Isotonic absorption. Generates and secretes NH_3, which enables the kidney to secrete more H^+.

PTH—inhibits Na^+/PO_4^{3-} cotransport → PO_4^{3-} excretion.

AT II—stimulates Na^+/H^+ exchange → ↑ Na^+, H_2O, and HCO_3^- reabsorption (permitting contraction alkalosis).

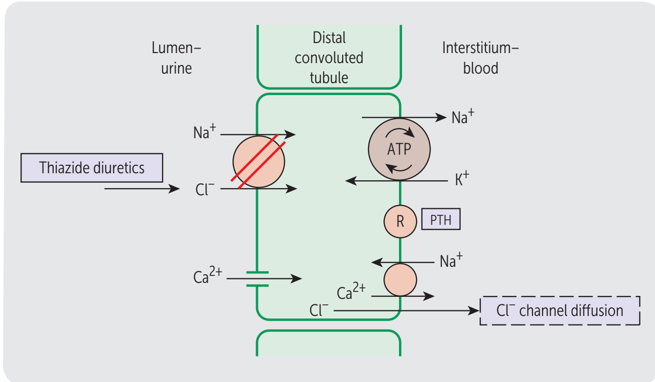
65–80% Na^+ reabsorbed.

Thin descending loop of Henle—passively reabsorbs H_2O via medullary hypertonicity (impermeable to Na^+). Concentrating segment. Makes urine hypertonic.



Thick ascending loop of Henle—reabsorbs Na^+, K^+, and Cl^-. Indirectly induces paracellular reabsorption of Mg^{2+} and Ca^{2+} through ⊕ lumen potential generated by K^+ backleak. Impermeable to H_2O. Makes urine less concentrated as it ascends.

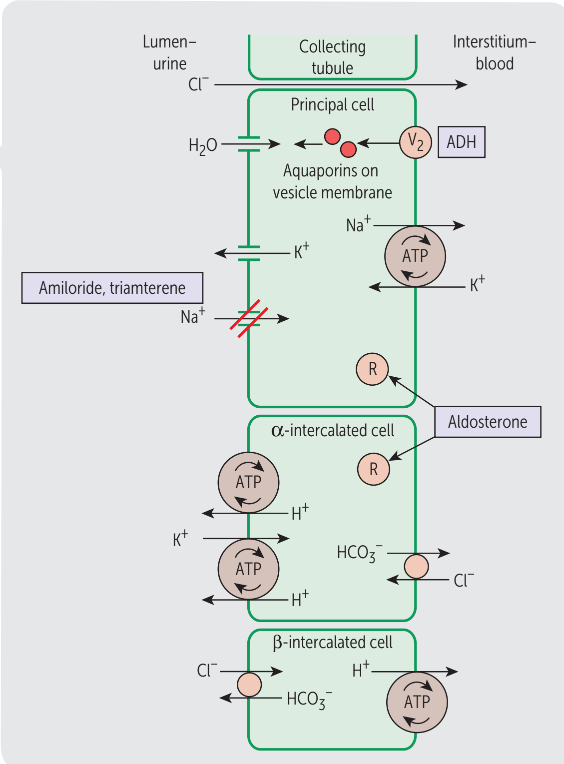
10–20% Na^+ reabsorbed.



Early DCT—reabsorbs Na^+, Cl^-. Makes urine fully dilute (hypotonic).

PTH—↑ Ca^{2+}/Na^+ exchange → Ca^{2+} reabsorption.

5–10% Na^+ reabsorbed.



Collecting tubule—reabsorbs Na^+ in exchange for secreting K^+ and H^+ (regulated by aldosterone).

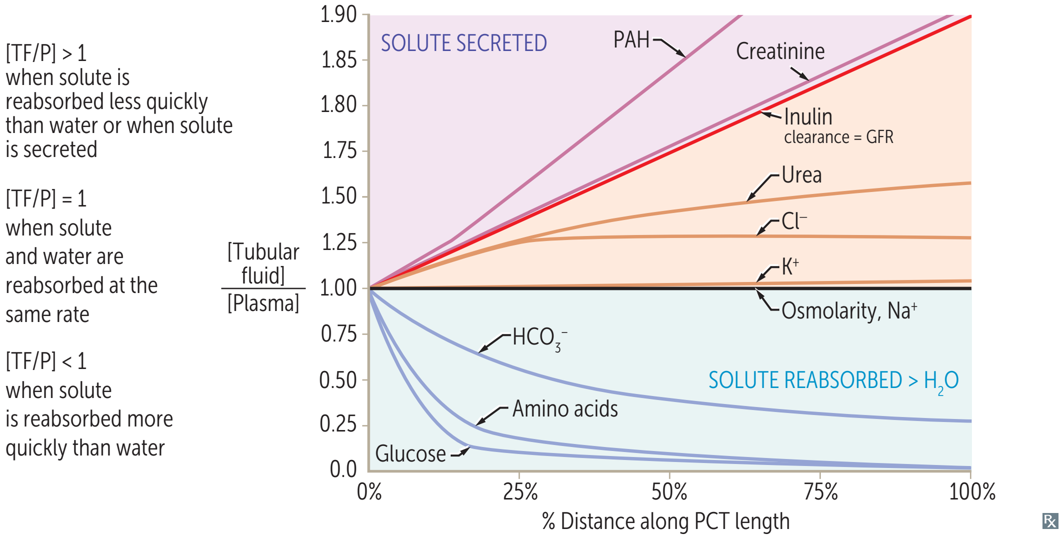
Aldosterone—acts on mineralocorticoid receptor → mRNA → protein synthesis. In principal cells: ↑ apical K^+ conductance, ↑ Na^+/K^+ pump, ↑ epithelial Na^+ channel (ENaC) activity → lumen negativity → K^+ secretion. In α-intercalated cells: lumen negativity → ↑ H^+ ATPase activity → ↑ H^+ secretion → ↑ HCO_3^-/Cl^- exchanger activity.

ADH—acts at V_2 receptor → insertion of aquaporin H_2O channels on apical side.

3–5% Na^+ reabsorbed.

Renal tubular defects	Fanconi syndrome is first (PCT), the rest are in alphabetic order.
Fanconi syndrome	Generalized reabsorptive defect in PCT. Associated with ↑ excretion of nearly all amino acids, glucose, HCO_3^-, and PO_4^{3-}. May result in metabolic acidosis (proximal renal tubular acidosis). Causes include hereditary defects (eg, Wilson disease, tyrosinemia, glycogen storage disease, cystinosis), ischemia, multiple myeloma, nephrotoxins/drugs (eg, ifosfamide, cisplatin, tenofovir, expired tetracyclines), lead poisoning.
Bartter syndrome	Reabsorptive defect in thick ascending loop of Henle. Affects $Na^+/K^+/2Cl^-$ cotransporter. Results in hypokalemia and metabolic alkalosis with hypercalciuria. Presents similarly to chronic loop diuretic use. Autosomal recessive.
Gitelman syndrome	Reabsorptive defect of NaCl in DCT. Leads to hypokalemia, hypomagnesemia, metabolic alkalosis, hypocalciuria. Similar to using life-long thiazide diuretics. Autosomal recessive. Less severe than Bartter syndrome.
Liddle syndrome	Gain of function mutation → ↑ Na^+ reabsorption in collecting tubules (↑ activity of Na^+ channel). Results in hypertension, hypokalemia, metabolic alkalosis, ↓ aldosterone. Presents like hyperaldosteronism, but aldosterone is nearly undetectable. Autosomal dominant. Treatment: amiloride.
Syndrome of Apparent Mineralocorticoid Excess	Hereditary deficiency of 11β-hydroxysteroid dehydrogenase, which normally converts cortisol (can activate mineralocorticoid receptors) to cortisone (inactive on mineralocorticoid receptors) in cells containing mineralocorticoid receptors. Excess cortisol in these cells from enzyme deficiency → ↑ mineralocorticoid receptor activity → hypertension, hypokalemia, metabolic alkalosis. Low serum aldosterone levels. Can acquire disorder from glycyrrhetinic acid (present in licorice), which blocks activity of 11β-hydroxysteroid dehydrogenase. Treatment: corticosteroids (exogenous corticosteroids ↓ endogenous cortisol production → ↓ mineralocorticoid receptor activation). Cortisol tries to be the SAME as aldosterone.

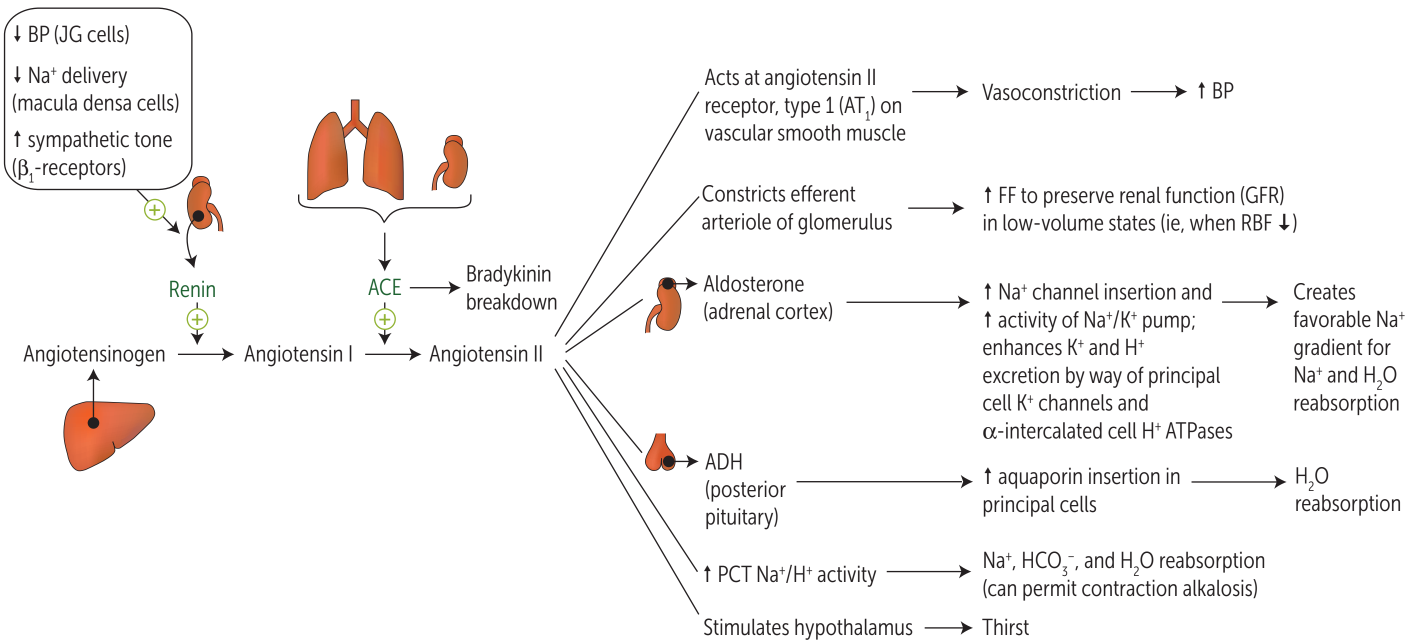
Relative concentrations along proximal convoluted tubules



Tubular inulin ↑ in concentration (but not amount) along the PCT as a result of water reabsorption.

Cl^- reabsorption occurs at a slower rate than Na^+ in early PCT and then matches the rate of Na^+ reabsorption more distally. Thus, its relative concentration ↑ before it plateaus.

Renin-angiotensin-aldosterone system



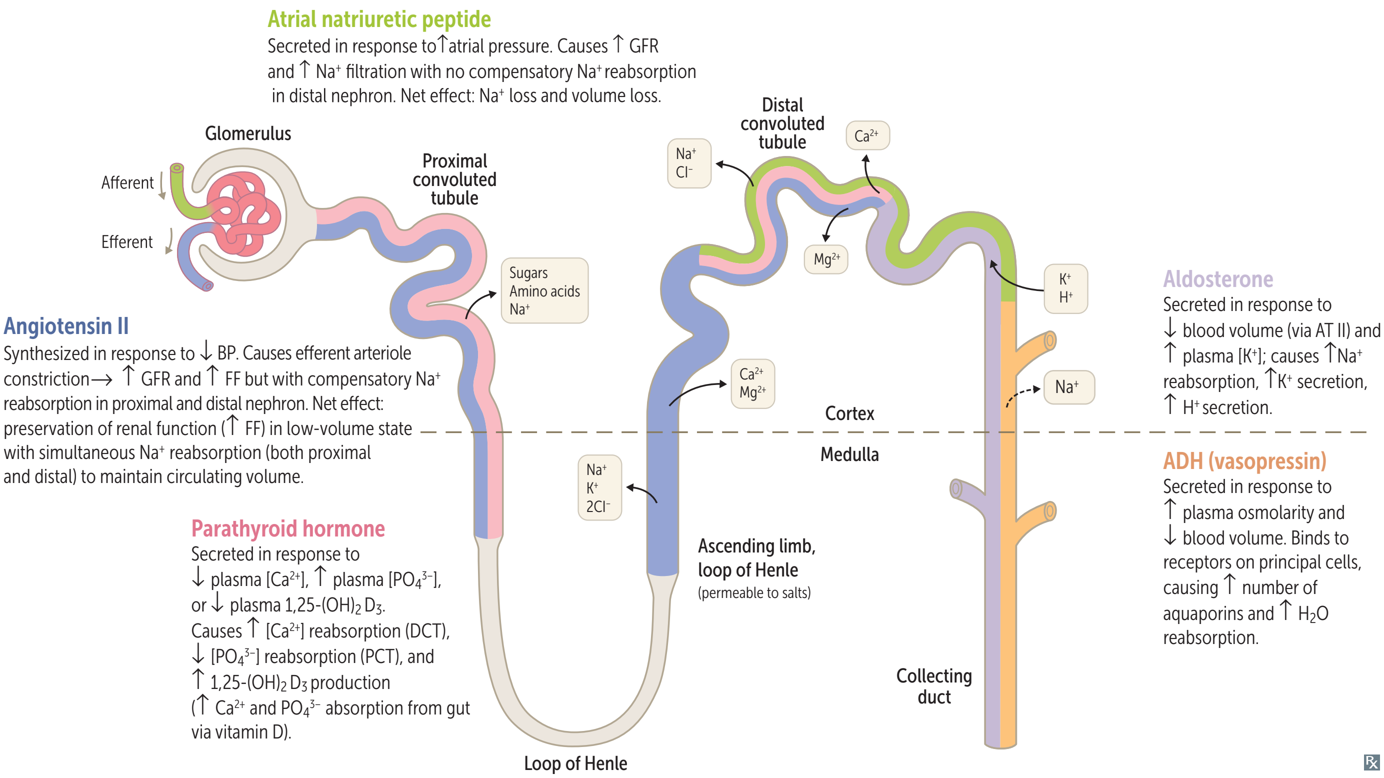
Renin	Secreted by JG cells in response to ↓ renal arterial pressure, ↑ renal sympathetic discharge (β_1 effect), and ↓ Na^+ delivery to macula densa cells.
AT II	Helps maintain blood volume and blood pressure. Affects baroreceptor function; limits reflex bradycardia, which would normally accompany its pressor effects.
ANP, BNP	Released from atria (ANP) and ventricles (BNP) in response to ↑ volume; may act as a "check" on renin-angiotensin-aldosterone system; relaxes vascular smooth muscle via cGMP → ↑ GFR, ↓ renin. Dilates afferent arteriole, constricts efferent arteriole, promotes natriuresis.
ADH	Primarily regulates osmolarity; also responds to low blood volume states.
Aldosterone	Primarily regulates ECF volume and Na^+ content; responds to low blood volume states. Responds to hyperkalemia by ↑ K^+ excretion.

Juxtaglomerular apparatus	Consists of mesangial cells, JG cells (modified smooth muscle of afferent arteriole) and the macula densa (NaCl sensor, part of DCT). JG cells secrete renin in response to ↓ renal blood pressure and ↑ sympathetic tone (β_1). Macula densa cells sense ↓ NaCl delivery to DCT → ↑ renin release → efferent arteriole vasoconstriction → ↑ GFR.	JGA maintains GFR via renin-angiotensin-aldosterone system. β-blockers can decrease BP by inhibiting β_1-receptors of the JGA → ↓ renin release.

Kidney endocrine functions

Erythropoietin	Released by interstitial cells in peritubular capillary bed in response to hypoxia.	Stimulates RBC proliferation in bone marrow. Erythropoietin often supplemented in chronic kidney disease.
Calciferol (vitamin D)	PCT cells convert 25-OH vitamin D_3 to 1,25-$(OH)_2$ vitamin D_3 (calcitriol, active form).	25-OH D_3 → 1,25-$(OH)_2$ D_3; 1α-hydroxylase; ⊕ PTH
Prostaglandins	Paracrine secretion vasodilates the afferent arterioles to ↑ RBF.	NSAIDs block renal-protective prostaglandin synthesis → constriction of afferent arteriole and ↓ GFR; this may result in acute renal failure in low renal blood flow states.
Dopamine	Secreted by PCT cells, promotes natriuresis. At low doses, dilates interlobular arteries, afferent arterioles, efferent arterioles → ↑ RBF, little or no change in GFR. At higher doses, acts as vasoconstrictor.	

Hormones acting on kidney



Potassium shifts

SHIFTS K^+ INTO CELL (CAUSING HYPOKALEMIA)	SHIFTS K^+ OUT OF CELL (CAUSING HYPERKALEMIA)
	Digitalis (blocks Na^+/K^+ ATPase)
Hypo-osmolarity	Hyper**O**smolarity
	Lysis of cells (eg, crush injury, rhabdomyolysis, tumor lysis syndrome)
Alkalosis	**A**cidosis
β-adrenergic agonist (↑ Na^+/K^+ ATPase)	**β**-blocker
Insulin (↑ Na^+/K^+ ATPase)	High blood **S**ugar (insulin deficiency)
Insulin shifts K^+ **in**to cells	**S**uccinylcholine (↑ risk in burns/muscle trauma)

Hyperkalemia? **DO LAβSS**

Electrolyte disturbances

ELECTROLYTE	LOW SERUM CONCENTRATION	HIGH SERUM CONCENTRATION
Na^+	Nausea and malaise, stupor, coma, seizures	Irritability, stupor, coma
K^+	U waves and flattened T waves on ECG, arrhythmias, muscle cramps, spasm, weakness	Wide QRS and peaked T waves on ECG, arrhythmias, muscle weakness
Ca^{2+}	Tetany, seizures, QT prolongation, twitching (Chvostek sign), spasm (Trousseau sign)	**Stones** (renal), **bones** (pain), **groans** (abdominal pain), **thrones** (↑ urinary frequency), **psychiatric overtones** (anxiety, altered mental status)
Mg^{2+}	Tetany, torsades de pointes, hypokalemia, hypocalcemia (when $[Mg^{2+}] < 1.2$ mg/dL)	↓ DTRs, lethargy, bradycardia, hypotension, cardiac arrest, hypocalcemia
PO_4^{3-}	Bone loss, osteomalacia (adults), rickets (children)	Renal stones, metastatic calcifications, hypocalcemia

Features of renal disorders

CONDITION	BLOOD PRESSURE	PLASMA RENIN	ALDOSTERONE	SERUM Mg^{2+}	URINE Ca^{2+}
Bartter syndrome	—	↑	↑		**↑**
Gitelman syndrome	—	↑	↑	↓	**↓**
Liddle syndrome	↑	↓	**↓**		
SIADH	—/↑	↓	↓		
Primary hyperaldosteronism (Conn syndrome)	↑	↓	**↑**		
Renin-secreting tumor	↑	**↑**	↑		

↑ ↓ = 1° disturbance.

Acid-base physiology

	pH	P_{CO_2}	$[HCO_3^-]$	COMPENSATORY RESPONSE
Metabolic acidosis	↓	↓	↓	Hyperventilation (immediate)
Metabolic alkalosis	↑	↑	↑	Hypoventilation (immediate)
Respiratory acidosis	↓	↑	↑	↑ renal $[HCO_3^-]$ reabsorption (delayed)
Respiratory alkalosis	↑	↓	↓	↓ renal $[HCO_3^-]$ reabsorption (delayed)

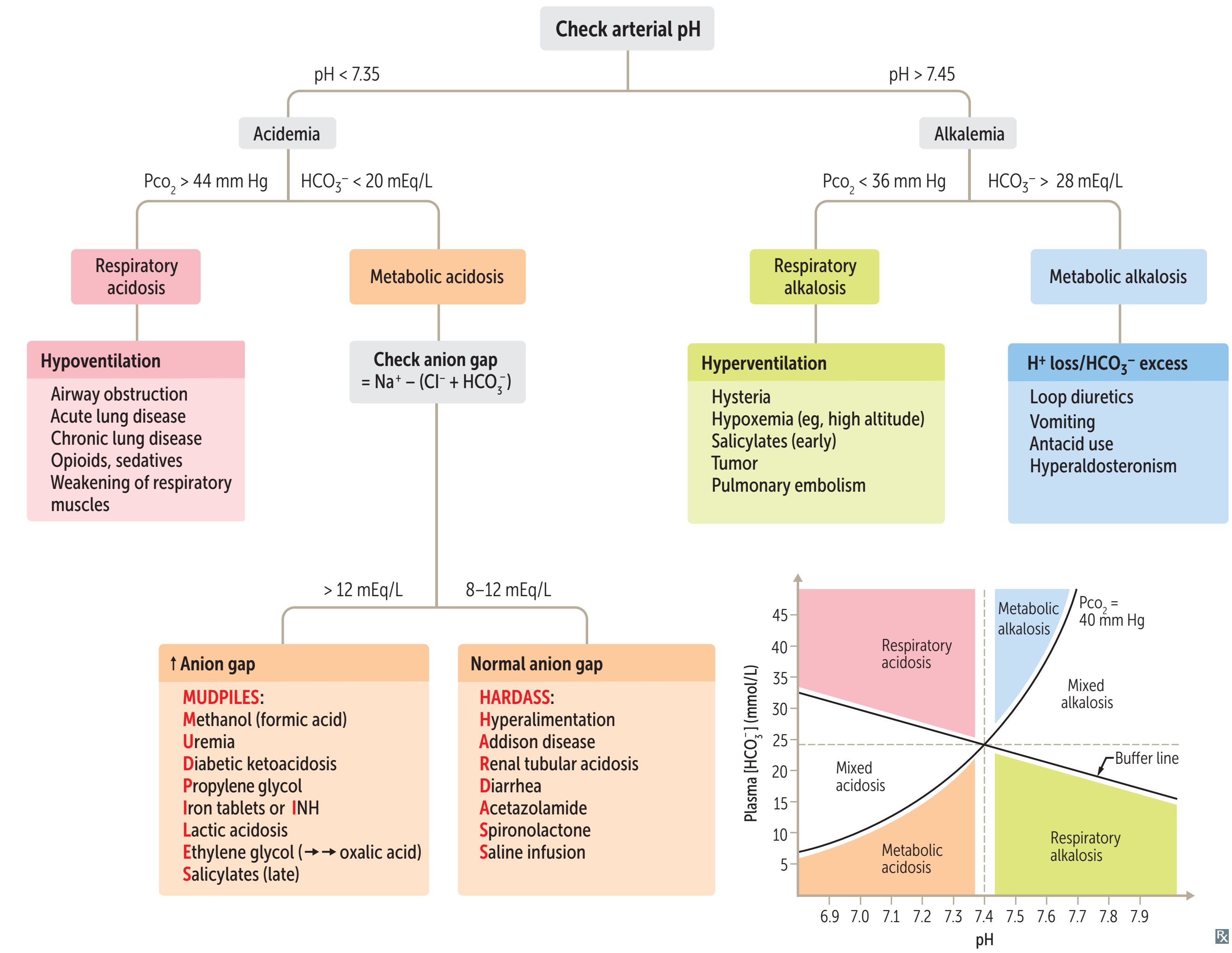
Key: ↑ ↓ = 1° disturbance; ↓ ↑ = compensatory response.

Henderson-Hasselbalch equation: $pH = 6.1 + \log \dfrac{[HCO_3^-]}{0.03\ P_{CO_2}}$

Predicted respiratory compensation for a simple metabolic acidosis can be calculated using the Winters formula. If measured P_{CO_2} > predicted P_{CO_2} → concomitant respiratory acidosis; if measured P_{CO_2} < predicted P_{CO_2} → concomitant respiratory alkalosis:

$$P_{CO_2} = 1.5\,[HCO_3^-] + 8 \pm 2$$

Acidosis and alkalosis

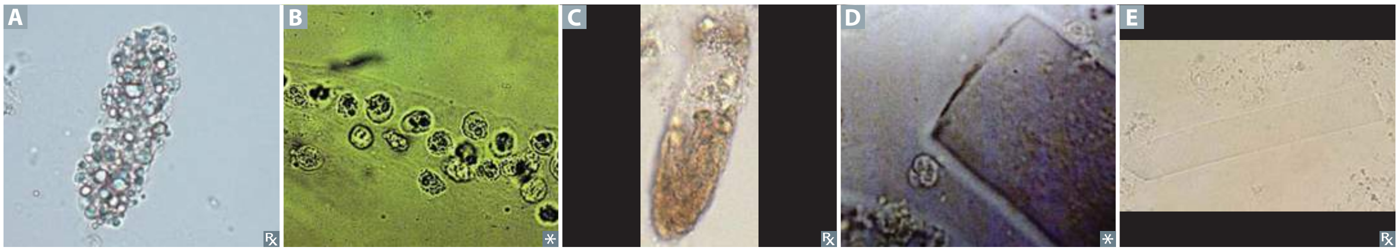


Renal tubular acidosis A disorder of the renal tubules that leads to normal anion gap (hyperchloremic) metabolic acidosis.

RTA TYPE	NOTES
Distal renal tubular acidosis (type 1)	Urine pH > 5.5. Defect in ability of α intercalated cells to secrete H^+ → no new HCO_3^- is generated → metabolic acidosis. Associated with **hypokalemia**, ↑ risk for calcium phosphate kidney stones (due to ↑ urine pH and ↑ bone turnover). Causes: amphotericin B toxicity, analgesic nephropathy, congenital anomalies (obstruction) of urinary tract.
Proximal renal tubular acidosis (type 2)	Urine pH < 5.5. Defect in PCT HCO_3^- reabsorption → ↑ excretion of HCO_3^- in urine and subsequent metabolic acidosis. Urine is acidified by α-intercalated cells in collecting tubule. Associated with **hypokalemia**, ↑ risk for hypophosphatemic rickets. Causes: Fanconi syndrome and carbonic anhydrase inhibitors.
Hyperkalemic renal tubular acidosis (type 4)	Urine pH < 5.5. Hypoaldosteronism → **hyperkalemia** → ↓ NH_3 synthesis in PCT → ↓ NH_4^+ excretion. Causes: ↓ aldosterone production (eg, diabetic hyporeninism, ACE inhibitors, ARBs, NSAIDs, heparin, cyclosporine, adrenal insufficiency) or aldosterone resistance (eg, K^+-sparing diuretics, nephropathy due to obstruction, TMP/SMX).

▸ RENAL—PATHOLOGY

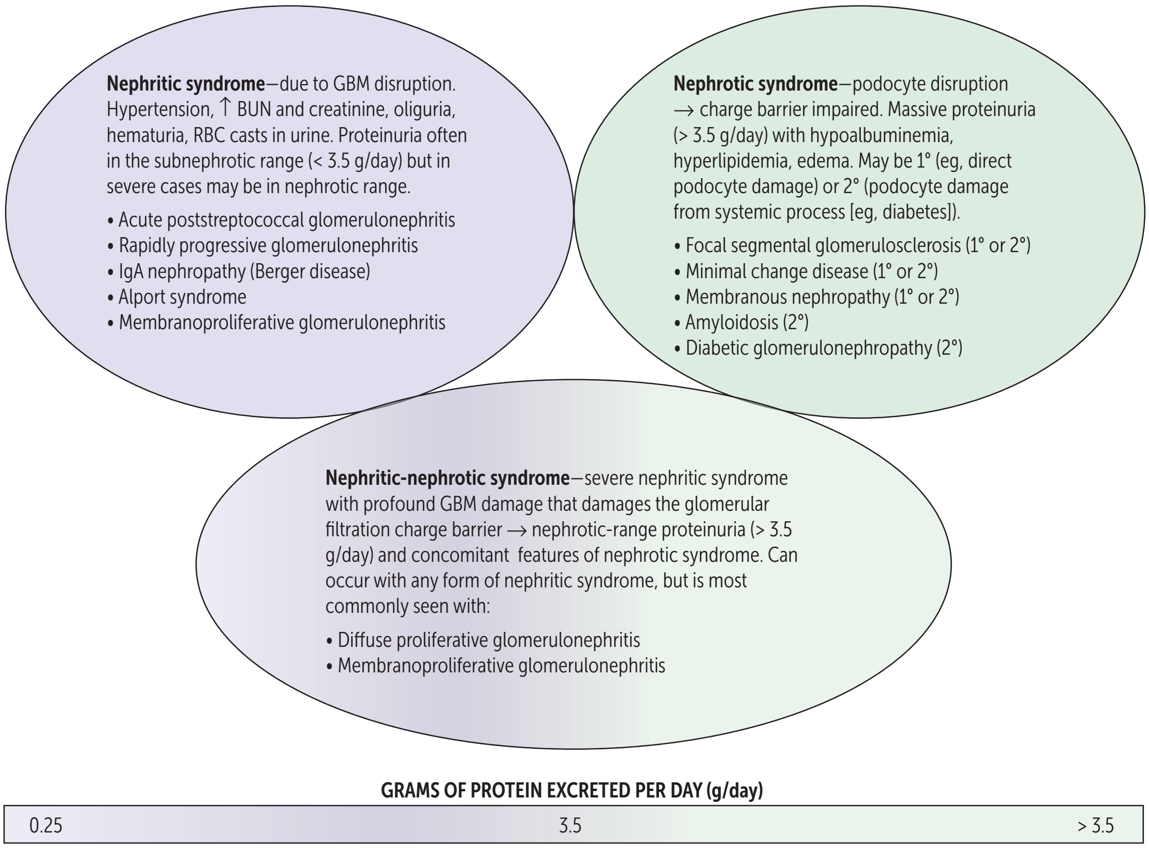
Casts in urine	Presence of casts indicates that hematuria/pyuria is of glomerular or renal tubular origin. Bladder cancer, kidney stones → hematuria, no casts. Acute cystitis → pyuria, no casts.
RBC casts A	Glomerulonephritis, malignant hypertension.
WBC casts B	Tubulointerstitial inflammation, acute pyelonephritis, transplant rejection.
Fatty casts ("oval fat bodies")	Nephrotic syndrome. Associated with "Maltese cross" sign.
Granular ("muddy brown") casts C	Acute tubular necrosis.
Waxy casts D	End-stage renal disease/chronic renal failure.
Hyaline casts E	Nonspecific, can be a normal finding, often seen in concentrated urine samples.



Nomenclature of glomerular disorders

TYPE	CHARACTERISTICS	EXAMPLE
Focal	< 50% of glomeruli are involved	Focal segmental glomerulosclerosis
Diffuse	> 50% of glomeruli are involved	Diffuse proliferative glomerulonephritis
Proliferative	Hypercellular glomeruli	Membranoproliferative glomerulonephritis
Membranous	Thickening of glomerular basement membrane (GBM)	Membranous nephropathy
Primary glomerular disease	1° disease of the kidney specifically impacting the glomeruli	Minimal change disease
Secondary glomerular disease	Systemic disease or disease of another organ system that also impacts the glomeruli	SLE, diabetic nephropathy

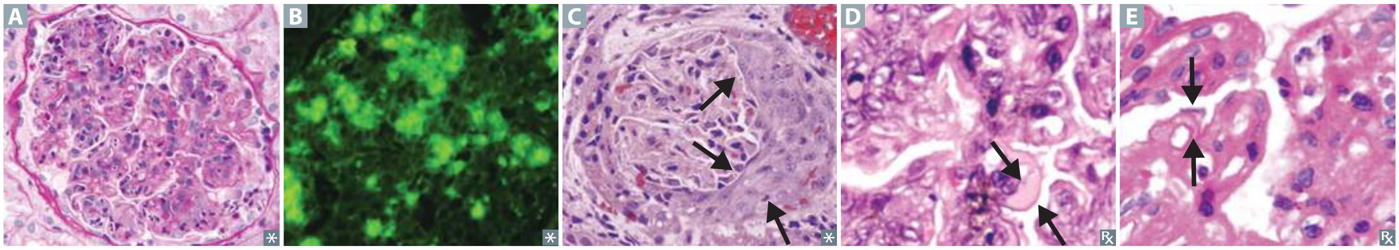
Glomerular diseases



Nephritic syndrome	NephrItic syndrome = Inflammatory process. When glomeruli are involved, leads to hematuria and RBC casts in urine. Associated with azotemia, oliguria, hypertension (due to salt retention), proteinuria.	
Acute poststreptococcal glomerulonephritis	LM—glomeruli enlarged and hypercellular A. IF—("starry sky") granular appearance ("lumpy-bumpy") B due to IgG, IgM, and C3 deposition along GBM and mesangium. EM—subepithelial immune complex (IC) humps.	Most frequently seen in children. Occurs ~ 2–4 weeks after group A streptococcal infection of pharynx or skin. Resolves spontaneously. Type III hypersensitivity reaction. Presents with peripheral and periorbital edema, cola-colored urine, hypertension. Positive strep titers/serologies, ↓ complement levels (C3) due to consumption.
Rapidly progressive (crescentic) glomerulonephritis	LM and IF—crescent moon shape C. Crescents consist of fibrin and plasma proteins (eg, C3b) with glomerular parietal cells, monocytes, macrophages. Several disease processes may result in this pattern, in particular:	Poor prognosis. Rapidly deteriorating renal function (days to weeks).
	▪ Goodpasture syndrome—type II hypersensitivity reaction; antibodies to GBM and alveolar basement membrane → linear IF	Hematuria/hemoptysis. Treatment: emergent plasmapheresis.
	▪ Granulomatosis with polyangiitis (Wegener)	PR3-ANCA/c-ANCA. Pauci-immune (no Ig/C3 deposition).
	▪ Microscopic Polyangiitis	MPO-ANCA/p-ANCA. Pauci-immune (no Ig/C3 deposition).
Diffuse proliferative glomerulonephritis	Often due to SLE or membranoproliferative glomerulonephritis. LM—"wire looping" of capillaries. EM—subendothelial and sometimes intramembranous IgG-based ICs often with C3 deposition. IF—granular.	A common cause of death in SLE (think "wire lupus"). DPGN and MPGN often present as nephrotic syndrome and nephritic syndrome concurrently.
IgA nephropathy (Berger disease)	LM—mesangial proliferation. EM—mesangial IC deposits. IF—IgA-based IC deposits in mesangium. Renal pathology of Henoch-Schönlein purpura.	Episodic gross hematuria that occurs concurrently with respiratory or GI tract infections (IgA is secreted by mucosal linings). Not to be confused with Buerger disease (thromboangiitis obliterans).

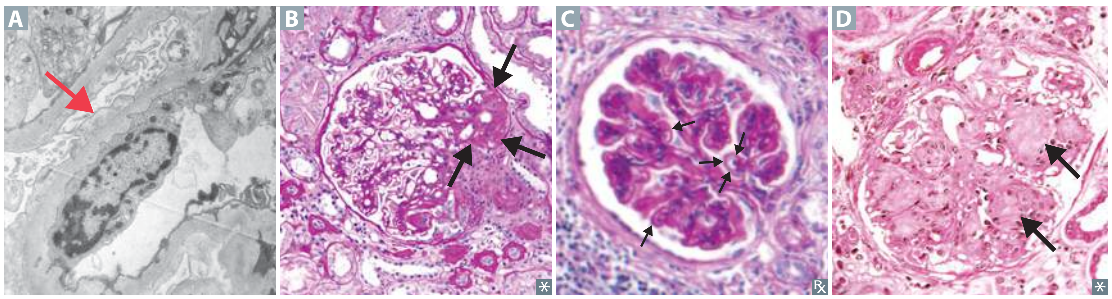
Nephritic syndrome *(continued)*

Alport syndrome	Mutation in type IV collagen → thinning and splitting of glomerular basement membrane. Most commonly X-linked dominant.	Eye problems (eg, retinopathy, lens dislocation), glomerulonephritis, sensorineural deafness; "can't see, can't pee, can't hear a bee." "Basket-weave" appearance on EM.
Membrano-proliferative glomerulonephritis	Type I—subendothelial immune complex (IC) deposits with granular IF; "tram-track" appearance on PAS stain D and H&E stain E due to GBM splitting caused by mesangial ingrowth. Type II—also called dense deposit disease.	MPGN is a nephritic syndrome that often copresents with nephrotic syndrome. Type I may be 2° to hepatitis B or C infection. May also be idiopathic. Type II is associated with C3 nephritic factor (IgG antibody that stabilizes C3 convertase → persistent complement activation → ↓ C3 levels).



LM = light microscopy; EM = electron microscopy; IF = immunofluorescence.

Nephrotic syndrome	NephrOtic syndrome—massive prOteinuria (> 3.5 g/day) with hypoalbuminemia, resulting edema, hyperlipidemia. Frothy urine with fatty casts. Due to podocyte damage disrupting glomerular filtration charge barrier. May be 1° (eg, direct sclerosis of podocytes) or 2° (systemic process [eg, diabetes] secondarily damages podocytes). Associated with hypercoagulable state (eg, thromboembolism) due to antithrombin (AT) III loss in urine and $\uparrow$ risk of infection (due to loss of immunoglobulins in urine and soft tissue compromise by edema). Severe nephritic syndrome may present with nephrotic syndrome features (nephritic-nephrotic syndrome) if damage to GBM is severe enough to damage charge barrier.	
Minimal change disease (lipoid nephrosis)	LM—normal glomeruli (lipid may be seen in PCT cells). IF $\ominus$. EM—effacement of foot processes A.	Most common cause of nephrotic syndrome in children. Often 1° (idiopathic) and may be triggered by recent infection, immunization, immune stimulus. Rarely, may be 2° to lymphoma (eg, cytokine-mediated damage). 1° disease has excellent response to corticosteroids.
Focal segmental glomerulosclerosis	LM—segmental sclerosis and hyalinosis B. IF—often $\ominus$, but may be $\oplus$ for nonspecific focal deposits of IgM, C3, C1. EM—effacement of foot process similar to minimal change disease.	Most common cause of nephrotic syndrome in African Americans and Hispanics. Can be 1° (idiopathic) or 2° to other conditions (eg, HIV infection, sickle cell disease, heroin abuse, massive obesity, interferon treatment, chronic kidney disease due to congenital malformations). 1° disease has inconsistent response to steroids. May progress to chronic renal disease.
Membranous nephropathy (membranous glomerulonephritis)	LM—diffuse capillary and GBM thickening C. IF—granular as a result of immune complex deposition. Nephrotic presentation of SLE. EM—"spike and dome" appearance with subepithelial deposits.	Most common cause of 1° nephrotic syndrome in Caucasian adults. Can be 1° (eg, antibodies to phospholipase A_2 receptor) or 2° to drugs (eg, NSAIDs, penicillamine, gold), infections (eg, HBV, HCV, syphilis), SLE, or solid tumors. 1° disease has poor response to steroids. May progress to chronic renal disease.
Amyloidosis	LM—Congo red stain shows apple-green birefringence under polarized light due to amyloid deposition in the mesangium.	Kidney is the most commonly involved organ (systemic amyloidosis). Associated with chronic conditions that predispose to amyloid deposition (eg, AL amyloid, AA amyloid).
Diabetic glomerulonephropathy	LM—mesangial expansion, GBM thickening, eosinophilic nodular glomerulosclerosis (Kimmelstiel-Wilson lesions, arrows in D).	Nonenzymatic glycosylation of GBM $\rightarrow \uparrow$ permeability, thickening. Nonenzymatic glycosylation of efferent arterioles (hyaline arteriosclerosis) $\rightarrow \uparrow$ GFR $\rightarrow$ mesangial expansion. Most common cause of end-stage renal disease in the United States.

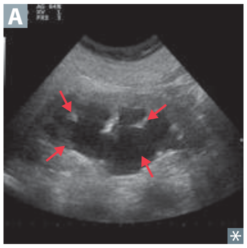


Kidney stones

Can lead to severe complications such as hydronephrosis, pyelonephritis. Presents with unilateral flank tenderness, colicky pain radiating to groin, hematuria. Treat and prevent by encouraging fluid intake.
Most common kidney stone presentation: calcium oxalate stone in patient with hypercalciuria and normocalcemia.

CONTENT	PRECIPITATES WITH	X-RAY FINDINGS	CT FINDINGS	URINE CRYSTAL	NOTES
Calcium	Calcium oxalate: hypocitraturia	Radiopaque	Radiopaque	Shaped like envelope A or dumbbell	Calcium stones most common (80%); calcium oxalate more common than calcium phosphate stones. Hypocitraturia often associated with ↓ urine pH. Can result from ethylene glycol (antifreeze) ingestion, vitamin C abuse, hypocitraturia, malabsorption (eg, Crohn disease). Treatment: thiazides, citrate, low-sodium diet.
	Calcium phosphate: ↑ pH	Radiopaque	Radiopaque	Wedge-shaped prism	Treatment: low-sodium diet, thiazides.
Ammonium magnesium phosphate	↑ pH	Radiopaque	Radiopaque	Coffin lid B	Also known as struvite; account for 15% of stones. Caused by infection with urease ⊕ bugs (eg, *Proteus mirabilis*, *Staphylococcus saprophyticus*, *Klebsiella*) that hydrolyze urea to ammonia → urine alkalinization. Commonly form staghorn calculi C. Treatment: eradication of underlying infection, surgical removal of stone.
Uric acid	↓ pH	RadiolUcent	Minimally visible	Rhomboid D or rosettes	About 5% of all stones. Risk factors: ↓ urine volume, arid climates, acidic pH. Strong association with hyperuricemia (eg, gout). Often seen in diseases with ↑ cell turnover (eg, leukemia). Treatment: alkalinization of urine, allopurinol.
Cystine	↓ pH	Radiolucent	Sometimes visible	Hexagonal E	Hereditary (autosomal recessive) condition in which Cystine-reabsorbing PCT transporter loses function, causing cystinuria. Transporter defect also results in poor reabsorption of Ornithine, Lysine, Arginine (COLA). Cystine is poorly soluble, thus stones form in urine. Usually begins in childhood. Can form staghorn calculi. Sodium cyanide nitroprusside test ⊕. "SIXtine" stones have SIX sides. Treatment: low sodium diet, alkalinization of urine, chelating agents if refractory.

Hydronephrosis



Distention/dilation of renal pelvis and calyces A. Usually caused by urinary tract obstruction (eg, renal stones, severe BPH, cervical cancer, injury to ureter); other causes include retroperitoneal fibrosis, vesicoureteral reflux. Dilation occurs proximal to site of pathology. Serum creatinine becomes elevated if obstruction is bilateral or if patient has only one kidney. Leads to compression and possible atrophy of renal cortex and medulla.

Renal cell carcinoma

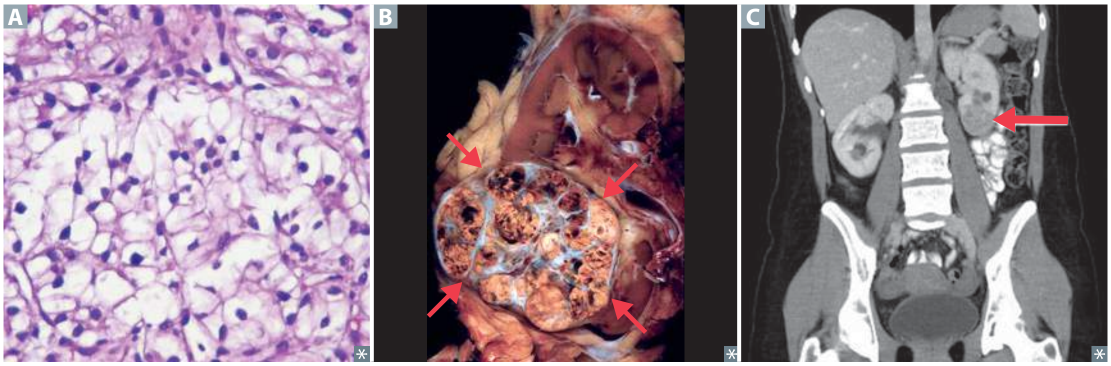
Originates from PCT cells → polygonal clear cells A filled with accumulated lipids and carbohydrates. Often golden-yellow B due to ↑ lipid content. Most common in men 50–70 years old. ↑ incidence with smoking and obesity. Manifests clinically with hematuria, palpable mass, 2° polycythemia, flank pain, fever, weight loss. Invades renal vein (may develop varicocele if left sided) then IVC and spreads hematogenously; metastasizes to lung and bone.

Treatment: surgery/ablation for localized disease. Immunotherapy (eg, aldesleukin) or targeted therapy for metastatic disease, rarely curative. Resistant to chemotherapy and radiation therapy.

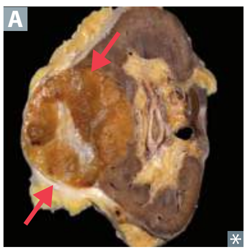
Most common 1° renal malignancy C.

Associated with gene deletion on chromosome 3 (sporadic or inherited as von Hippel-Lindau syndrome). RCC = 3 letters = chromosome 3.

Associated with paraneoplastic syndromes (eg, ectopic EPO, ACTH, PTHrP, renin).



Renal oncocytoma

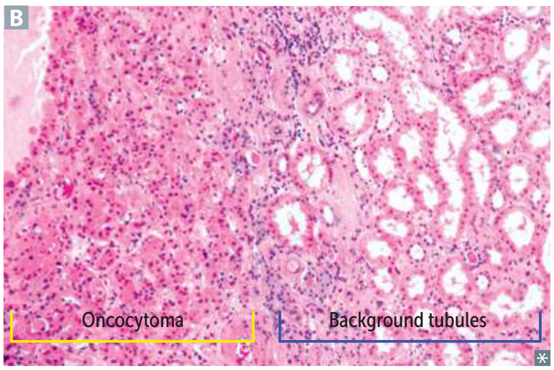


Benign epithelial cell tumor arising from collecting ducts (arrows in A point to well-circumscribed mass with central scar).

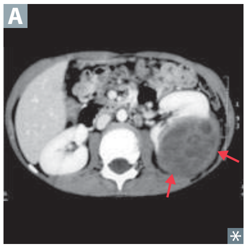
Large eosinophilic cells with abundant mitochondria without perinuclear clearing B (vs chromophobe renal cell carcinoma).

Presents with painless hematuria, flank pain, abdominal mass.

Often resected to exclude malignancy (eg, renal cell carcinoma).



Nephroblastoma (Wilms tumor)



Most common renal malignancy of early childhood (ages 2–4). Contains embryonic glomerular structures. Presents with large, palpable, unilateral flank mass A and/or hematuria.

"Loss of function" mutations of tumor suppressor genes *WT1* or *WT2* on chromosome 11.

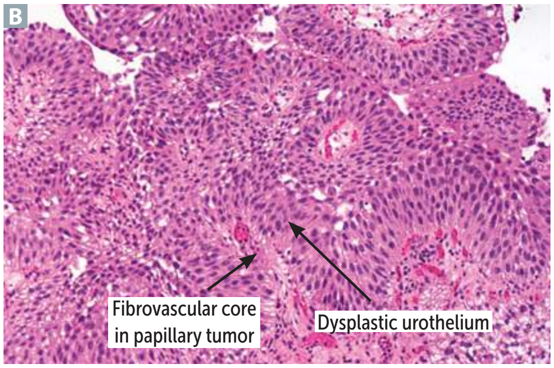
May be a part of several syndromes:

- **WAGR** complex: **W**ilms tumor, **A**niridia (absence of iris), **G**enitourinary malformations, mental **R**etardation/intellectual disability (*WT1* deletion)
- Denys-Drash: Wilms tumor, early-onset nephrotic syndrome, male pseudohermaphroditism (*WT1* mutation)
- Beckwith-Wiedemann: Wilms tumor, macroglossia, organomegaly, hemihyperplasia (*WT2* mutation)

Transitional cell carcinoma

Most common tumor of urinary tract system (can occur in renal calyces, renal pelvis, ureters, and bladder) A B. Can be suggested by painless hematuria (no casts).

Associated with problems in your **P**ee **SAC**: **P**henacetin, **S**moking, **A**niline dyes, and **C**yclophosphamide.



Squamous cell carcinoma of the bladder

Chronic irritation of urinary bladder → squamous metaplasia → dysplasia and squamous cell carcinoma.

Risk factors include *Schistosoma haematobium* infection (Middle East), chronic cystitis, smoking, chronic nephrolithiasis. Presents with painless hematuria.

Urinary incontinence

Stress incontinence	Outlet incompetence (urethral hypermobility or intrinsic sphincteric deficiency) → leak with ↑ intra-abdominal pressure (eg, sneezing, lifting). ↑ risk with obesity, vaginal delivery, prostate surgery. ⊕ bladder stress test (directly observed leakage from urethra upon coughing or Valsalva maneuver). Treatment: pelvic floor muscle strengthening (Kegel) exercises, weight loss, pessaries.
Urgency incontinence	Overactive bladder (detrusor instability) → leak with urge to void immediately. Treatment: Kegel exercises, bladder training (timed voiding, distraction or relaxation techniques), antimuscarinics (eg, oxybutynin).
Mixed incontinence	Features of both stress and urgency incontinence.
Overflow incontinence	Incomplete emptying (detrusor underactivity or outlet obstruction) → leak with overfilling. ↑ post-void residual (urinary retention) on catheterization or ultrasound. Treatment: catheterization, relieve obstruction (eg, α-blockers for BPH).

Urinary tract infection (acute bacterial cystitis)

Inflammation of urinary bladder. Presents as suprapubic pain, dysuria, urinary frequency, urgency. Systemic signs (eg, high fever, chills) are usually absent.

Risk factors include female gender (short urethra), sexual intercourse ("honeymoon cystitis"), indwelling catheter, diabetes mellitus, impaired bladder emptying.

Causes:

- *E coli* (most common).
- *Staphylococcus saprophyticus*—seen in sexually active young women (*E coli* is still more common in this group).
- *Klebsiella*.
- *Proteus mirabilis*—urine has ammonia scent.

Lab findings: ⊕ leukocyte esterase. ⊕ nitrites (indicate gram ⊖ organisms). Sterile pyuria and ⊖ urine cultures suggest urethritis by *Neisseria gonorrhoeae* or *Chlamydia trachomatis*.

Pyelonephritis

Acute pyelonephritis

Neutrophils infiltrate renal interstitium A. Affects cortex with relative sparing of glomeruli/vessels. Presents with fevers, flank pain (costovertebral angle tenderness), nausea/vomiting, chills.

Causes include ascending UTI (*E coli* is most common), hematogenous spread to kidney. Presents with WBCs in urine +/– WBC casts. CT would show striated parenchymal enhancement B.

Risk factors include indwelling urinary catheter, urinary tract obstruction, vesicoureteral reflux, diabetes mellitus, pregnancy.

Complications include chronic pyelonephritis, renal papillary necrosis, perinephric abscess, urosepsis.

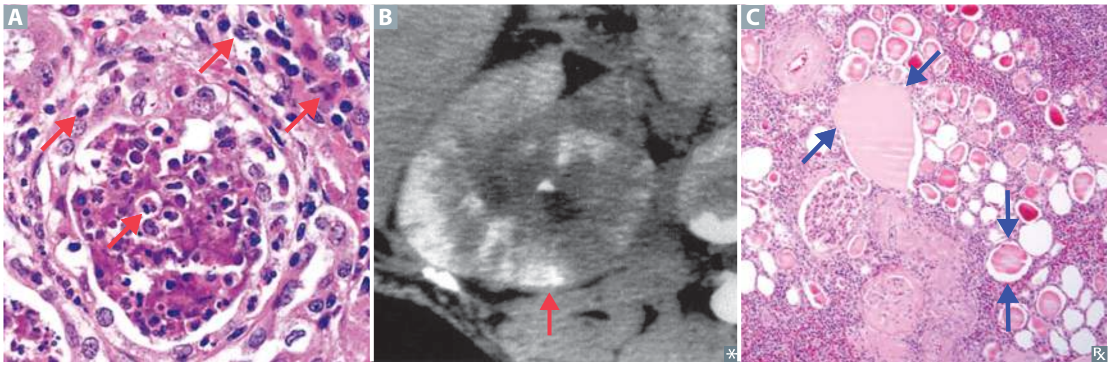
Treatment: antibiotics.

Chronic pyelonephritis

The result of recurrent episodes of acute pyelonephritis. Typically requires predisposition to infection such as vesicoureteral reflux or chronically obstructing kidney stones.

Coarse, asymmetric corticomedullary scarring, blunted calyx. Tubules can contain eosinophilic casts resembling thyroid tissue C (thyroidization of kidney).

Xanthogranulomatous pyelonephritis—rare; grossly orange nodules that can mimic tumor nodules; characterized by widespread kidney damage due to granulomatous tissue containing foamy macrophages.



Diffuse cortical necrosis

Acute generalized cortical infarction of both kidneys. Likely due to a combination of vasospasm and DIC.

Associated with obstetric catastrophes (eg, abruptio placentae), septic shock.

Renal osteodystrophy

Hypocalcemia, hyperphosphatemia, and failure of vitamin D hydroxylation associated with chronic renal disease → 2° hyperparathyroidism. Hyperphosphatemia also independently ↓ serum Ca^{2+} by causing tissue calcifications, whereas ↓ 1,25-$(OH)_2$ D_3 → ↓ intestinal Ca^{2+} absorption. Causes subperiosteal thinning of bones.

Acute kidney injury (acute renal failure)

Acute kidney injury is defined as an abrupt decline in renal function as measured by ↑ creatinine and ↑ BUN or by oliguria/anuria.

Prerenal azotemia

Due to ↓ RBF (eg, hypotension) → ↓ GFR. Na^+/H_2O and BUN retained by kidney in an attempt to conserve volume → ↑ BUN/creatinine ratio (BUN is reabsorbed, creatinine is not) and ↓ FE_{Na}.

Intrinsic renal failure

Generally due to acute tubular necrosis or ischemia/toxins; less commonly due to acute glomerulonephritis (eg, RPGN, hemolytic uremic syndrome) or acute interstitial nephritis.

In ATN, patchy necrosis → debris obstructing tubule and fluid backflow across necrotic tubule → ↓ GFR. Urine has epithelial/granular casts. BUN reabsorption is impaired → ↓ BUN/creatinine ratio and ↑ FE_{Na}.

Postrenal azotemia

Due to outflow obstruction (stones, BPH, neoplasia, congenital anomalies). Develops only with bilateral obstruction.

	Prerenal	Intrinsic renal	Postrenal
Urine osmolality (mOsm/kg)	> 500	< 350	< 350
Urine Na^+ (mEq/L)	< 20	> 40	> 40
FE_{Na}	< 1%	> 2%	< 1% (mild) > 2% (severe)
Serum BUN/Cr	> 20	< 15	Varies

Consequences of renal failure

Inability to make urine and excrete nitrogenous wastes.

Consequences (**MAD HUNGER**):

- **M**etabolic **A**cidosis
- **D**yslipidemia (especially ↑ triglycerides)
- **H**yperkalemia
- **U**remia—clinical syndrome marked by ↑ BUN:
  - Nausea and anorexia
  - Pericarditis
  - Asterixis
  - Encephalopathy
  - Platelet dysfunction
- **N**a^+/H_2O retention (HF, pulmonary edema, hypertension)
- **G**rowth retardation and developmental delay
- **E**rythropoietin failure (anemia)
- **R**enal osteodystrophy

2 forms of renal failure: acute (eg, ATN) and chronic (eg, hypertension, diabetes mellitus, congenital anomalies).

Acute interstitial nephritis (tubulointerstitial nephritis)

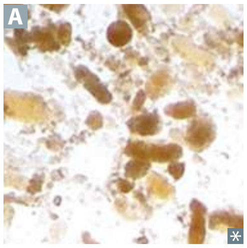
Acute interstitial renal inflammation. Pyuria (classically eosinophils) and azotemia occurring after administration of drugs that act as haptens, inducing hypersensitivity (eg, diuretics, penicillin derivatives, proton pump inhibitors, sulfonamides, rifampin, NSAIDs). Less commonly may be 2° to other processes such as systemic infections (eg, mycoplasma) or autoimmune diseases (eg, Sjögren syndrome, SLE, sarcoidosis).

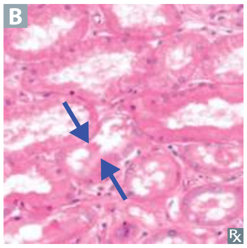
Associated with fever, rash, hematuria, and costovertebral angle tenderness, but can be asymptomatic.

Remember these P's:

- Pee (diuretics)
- Pain-free (NSAIDs)
- Penicillins and cephalosporins
- Proton pump inhibitors
- RifamPin

Acute tubular necrosis





Most common cause of acute kidney injury in hospitalized patients. Spontaneously resolves in many cases. Can be fatal, especially during initial oliguric phase. ↑ FENa.

Key finding: granular ("muddy brown") casts A.

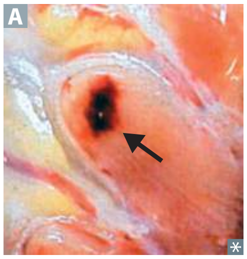
3 stages:

1. Inciting event
2. Maintenance phase—oliguric; lasts 1–3 weeks; risk of hyperkalemia, metabolic acidosis, uremia
3. Recovery phase—polyuric; BUN and serum creatinine fall; risk of hypokalemia

Can be caused by ischemic or nephrotoxic injury:

- Ischemic—2° to ↓ renal blood flow (eg, hypotension, shock, sepsis, hemorrhage, HF). Results in death of tubular cells that may slough into tubular lumen B (PCT and thick ascending limb are highly susceptible to injury).
- Nephrotoxic—2° to injury resulting from toxic substances (eg, aminoglycosides, radiocontrast agents, lead, cisplatin, ethylene glycol), crush injury (myoglobinuria), hemoglobinuria. PCT is particularly susceptible to injury.

Renal papillary necrosis



Sloughing of necrotic renal papillae A → gross hematuria and proteinuria. May be triggered by recent infection or immune stimulus. Associated with sickle cell disease or trait, acute pyelonephritis, NSAIDs, diabetes mellitus.

SAAD papa with papillary necrosis:

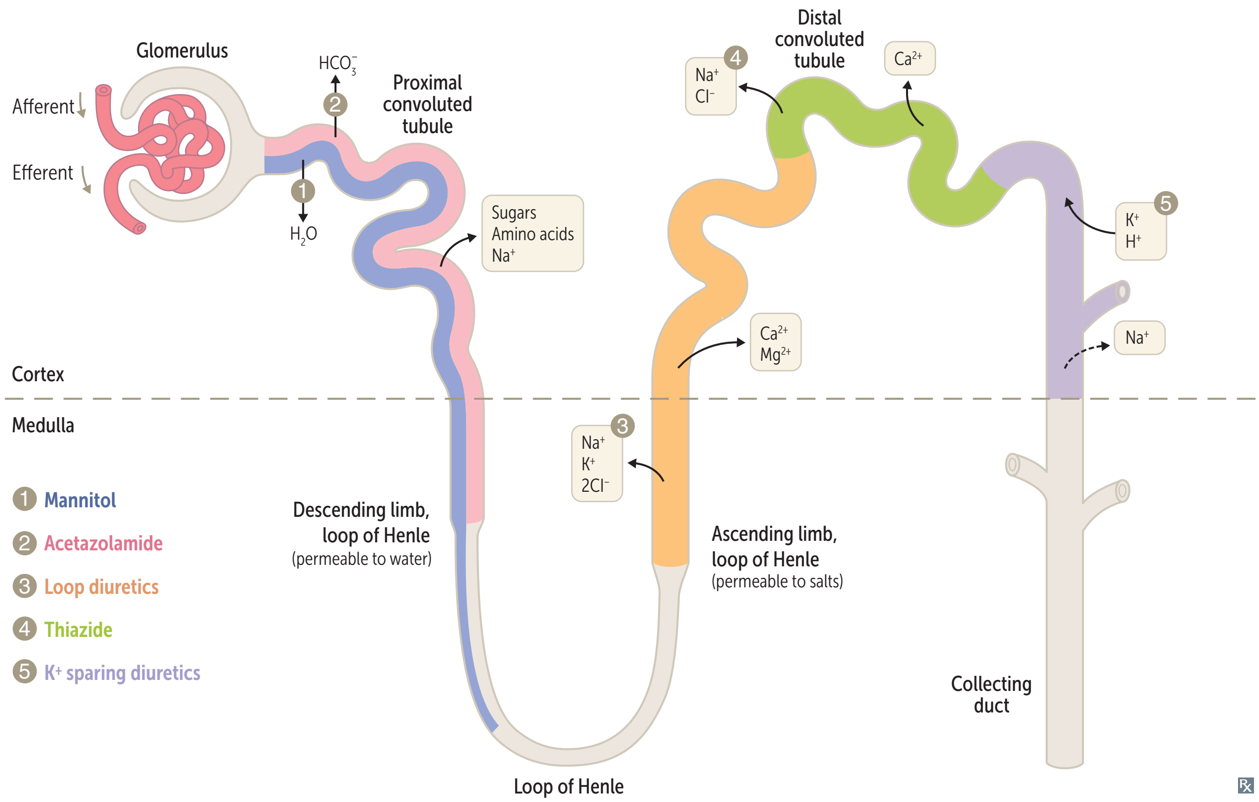
Sickle cell disease or trait
Acute pyelonephritis
Analgesics (NSAIDs)
Diabetes mellitus

Renal cyst disorders

Autosomal dominant polycystic kidney disease	Numerous cysts in cortex and medulla A causing bilateral enlarged kidneys ultimately destroy kidney parenchyma. Presents with flank pain, hematuria, hypertension, urinary infection, progressive renal failure in ~ 50% of individuals. Mutation in *PKD1* (85% of cases, chromosome 16) or *PKD2* (15% of cases, chromosome 4). Death from complications of chronic kidney disease or hypertension (caused by ↑ renin production). Associated with berry aneurysms, mitral valve prolapse, benign hepatic cysts, diverticulosis. Treatment: ACE inhibitors or ARBs.
Autosomal recessive polycystic kidney disease	Cystic dilation of collecting ducts B. Often presents in infancy. Associated with congenital hepatic fibrosis. Significant oliguric renal failure in utero can lead to Potter sequence. Concerns beyond neonatal period include systemic hypertension, progressive renal insufficiency, and portal hypertension from congenital hepatic fibrosis.
Medullary cystic disease	Inherited disease causing tubulointerstitial fibrosis and progressive renal insufficiency with inability to concentrate urine. Medullary cysts usually not visualized; shrunken kidneys on ultrasound. Poor prognosis.
Simple vs complex renal cysts	Simple cysts are filled with ultrafiltrate (anechoic on ultrasound C). Very common and account for majority of all renal masses. Found incidentally and typically asymptomatic. Complex cysts, including those that are septated, enhanced, or have solid components on imaging require follow-up or removal due to risk of renal cell carcinoma.

▶ RENAL—PHARMACOLOGY

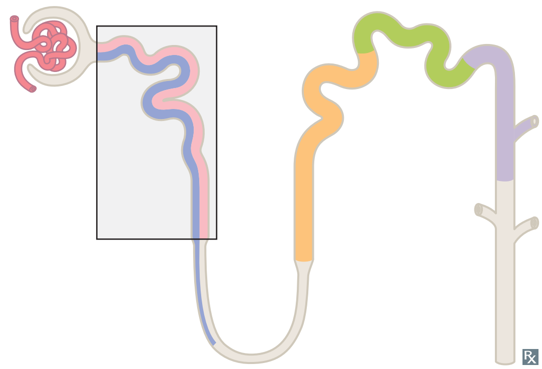
Diuretics site of action



Mannitol

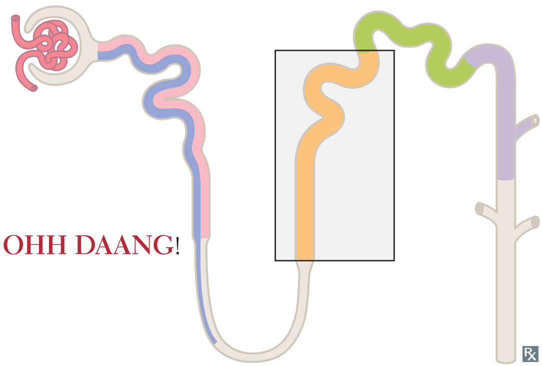
MECHANISM	Osmotic diuretic. ↑ tubular fluid osmolarity → ↑ urine flow, ↓ intracranial/intraocular pressure.
CLINICAL USE	Drug overdose, elevated intracranial/intraocular pressure.
ADVERSE EFFECTS	Pulmonary edema, dehydration. Contraindicated in anuria, HF.

Acetazolamide

MECHANISM	Carbonic anhydrase inhibitor. Causes self-limited $NaHCO_3$ diuresis and ↓ total body HCO_3^- stores.	
CLINICAL USE	Glaucoma, urinary alkalinization, metabolic alkalosis, altitude sickness, pseudotumor cerebri.	
ADVERSE EFFECTS	Proximal renal tubular acidosis, paresthesias, NH_3 toxicity, sulfa allergy, hypokalemia.	"ACID"azolamide causes ACIDosis.

Loop diuretics

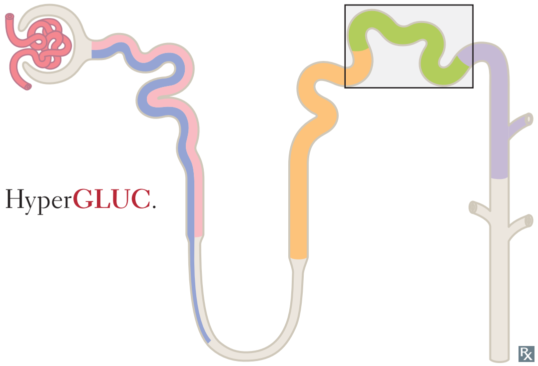
Furosemide, bumetanide, torsemide

MECHANISM	Sulfonamide loop diuretics. Inhibit cotransport system ($Na^+/K^+/2Cl^-$) of thick ascending limb of loop of Henle. Abolish hypertonicity of medulla, preventing concentration of urine. Stimulate PGE release (vasodilatory effect on afferent arteriole); inhibited by NSAIDs. ↑ Ca^{2+} excretion. Loops Lose Ca^{2+}.	
CLINICAL USE	Edematous states (HF, cirrhosis, nephrotic syndrome, pulmonary edema), hypertension, hypercalcemia.	
ADVERSE EFFECTS	Ototoxicity, Hypokalemia, Hypomagnesemia, Dehydration, Allergy (sulfa), metabolic Alkalosis, Nephritis (interstitial), Gout.	OHH DAANG!

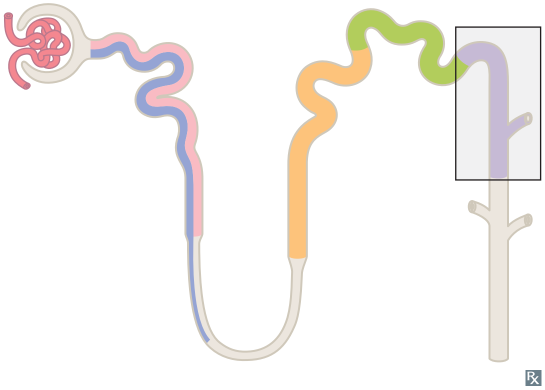
Ethacrynic acid

MECHANISM	Nonsulfonamide inhibitor of cotransport system ($Na^+/K^+/2Cl^-$) of thick ascending limb of loop of Henle.	
CLINICAL USE	Diuresis in patients allergic to sulfa drugs.	
ADVERSE EFFECTS	Similar to furosemide, but more ototoxic.	Loop earrings hurt your ears.

Thiazide diuretics	Hydrochlorothiazide, chlorthalidone, metolazone.	
MECHANISM	Inhibit NaCl reabsorption in early DCT → ↓ diluting capacity of nephron. ↓ Ca^{2+} excretion.	HyperGLUC.
CLINICAL USE	Hypertension, HF, idiopathic hypercalciuria, nephrogenic diabetes insipidus, osteoporosis.	
ADVERSE EFFECTS	Hypokalemic metabolic alkalosis, hyponatremia, hyperGlycemia, hyperLipidemia, hyperUricemia, hyperCalcemia. Sulfa allergy.	



Potassium-sparing diuretics	Spironolactone and eplerenone; Triamterene, and Amiloride.	The K^+ STAys.
MECHANISM	Spironolactone and eplerenone are competitive aldosterone receptor antagonists in cortical collecting tubule. Triamterene and amiloride act at the same part of the tubule by blocking Na^+ channels in the cortical collecting tubule.	
CLINICAL USE	Hyperaldosteronism, K^+ depletion, HF, hepatic ascites (spironolactone), nephrogenic DI (amiloride), antiandrogen.	
ADVERSE EFFECTS	Hyperkalemia (can lead to arrhythmias), endocrine effects with spironolactone (eg, gynecomastia, antiandrogen effects).	



Diuretics: electrolyte changes

Urine NaCl	↑ with all diuretics (strength varies based on potency of diuretic effect). Serum NaCl may decrease as a result.
Urine K^+	↑ especially with loop and thiazide diuretics. Serum K^+ may decrease as a result.
Blood pH	↓ (**acidemia**): carbonic anhydrase inhibitors: ↓ HCO_3^- reabsorption. K^+ sparing: aldosterone blockade prevents K^+ secretion and H^+ secretion. Additionally, hyperkalemia leads to K^+ entering all cells (via H^+/K^+ exchanger) in exchange for H^+ exiting cells. ↑ (**alkalemia**): loop diuretics and thiazides cause alkalemia through several mechanisms: ▪ Volume contraction → ↑ AT II → ↑ Na^+/H^+ exchange in PCT → ↑ HCO_3^- reabsorption ("contraction alkalosis") ▪ K^+ loss leads to K^+ exiting all cells (via H^+/K^+ exchanger) in exchange for H^+ entering cells ▪ In low K^+ state, H^+ (rather than K^+) is exchanged for Na^+ in cortical collecting tubule → alkalosis and "paradoxical aciduria"
Urine Ca^{2+}	↑ with loop diuretics: ↓ paracellular Ca^{2+} reabsorption → hypocalcemia. ↓ with thiazides: enhanced Ca^{2+} reabsorption.

Angiotensin-converting enzyme inhibitors

Captopril, enalapril, lisinopril, ramipril.

MECHANISM	Inhibit ACE → ↓ AT II → ↓ GFR by preventing constriction of efferent arterioles. ↑ renin due to loss of negative feedback. Inhibition of ACE also prevents inactivation of bradykinin, a potent vasodilator.	
CLINICAL USE	Hypertension, HF (↓ mortality), proteinuria, diabetic nephropathy. Prevent unfavorable heart remodeling as a result of chronic hypertension.	In chronic kidney disease (eg, diabetic nephropathy), ↓ intraglomerular pressure, slowing GBM thickening.
ADVERSE EFFECTS	**C**ough, **A**ngioedema (due to ↑ bradykinin; contraindicated in C1 esterase inhibitor deficiency), **T**eratogen (fetal renal malformations), ↑ **C**reatinine (↓ GFR), **H**yperkalemia, and **H**ypotension. Used with caution in bilateral renal artery stenosis because ACE inhibitors will further ↓ GFR → renal failure.	Captopril's **CATCHH**.

Angiotensin II receptor blockers

Losartan, candesartan, valsartan.

MECHANISM	Selectively block binding of angiotensin II to AT_1 receptor. Effects similar to ACE inhibitors, but ARBs do not increase bradykinin.
CLINICAL USE	Hypertension, HF, proteinuria, or chronic kidney disease (eg, diabetic nephropathy) with intolerance to ACE inhibitors (eg, cough, angioedema).
ADVERSE EFFECTS	Hyperkalemia, ↓ GFR, hypotension; teratogen.

Aliskiren

MECHANISM	Direct renin inhibitor, blocks conversion of angiotensinogen to angiotensin I.
CLINICAL USE	Hypertension.
ADVERSE EFFECTS	Hyperkalemia, ↓ GFR, hypotension, angioedema. Relatively contraindicated in patients already taking ACE inhibitors or ARBs.

NOTES